

Supplementary Material

When More Motion Modeling Hurts: Causal Latency Compensation for Streaming Object Detection in Automated Driving

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1 Analysis closure

The final analysis closure contained 100 valid detector-log scientific units and four shard-summary rows. There were no duplicate or missing scientific units. RT-DETR-L contained 50 prediction, 50 target, and 50 tracker outputs. The provenance audit passed. The core leakage audit found no future-information flags in deployable predictions. Twenty closure tests passed.

2 Tracker baseline results

Table 1: Normalized center error at 300 ms in high-yaw held-out logs.

Detector	Stale	Const. vel.	SORT	ByteTrack	OC-SORT	B3	B5
YOLO11n	.01826	.01279	.01924	.01513	.01219	.00700	.00915
YOLO11s	.01689	.01185	.01867	.01418	.01103	.00584	.00765
RT-DETR-L	.02001	.01547	.02193	.01968	.01392	.00851	.01018

3 Full-frame primary endpoint

Table 2: Pooled full-frame metrics at 300 ms in high-yaw logs.

Detector	Model	AP _{50:95}	AP ₇₅	Recall	FP/frame
YOLO11n	B3	.04046	.01818	.16683	3.5875
YOLO11n	B5	.03526	.01307	.16555	3.7011
YOLO11s	B3	.03964	.01582	.20951	4.9897
YOLO11s	B5	.03638	.01168	.20195	5.2212
RT-DETR-L	B3	.04661	.02175	.27261	14.0206
RT-DETR-L	B5	.04685	.01738	.25888	14.4438

Paired log-level AP_{50:95} intervals included zero for all three detectors. Paired AP₇₅ differences favored B3 for YOLO11n and YOLO11s and were borderline for RT-DETR-L. Recall and false-positive contrasts favored B3 across detectors.

4 Sensitivity results

Table 3: Selected sensitivity strata. Negative relative differences favor B5.

Stratum	Logs	Comparisons	Relative difference	Logs favoring B5
100 ms	50	110,377	2.58%	38%
200 ms	50	108,634	8.77%	28%
300 ms	50	106,650	15.50%	28%
400 ms	50	104,618	23.03%	27%
500 ms	50	102,557	30.40%	26%
Distance 0–20 m	50	98,802	−8.28%	69%
Distance 20–40 m	50	185,837	7.55%	40%
Distance > 40 m	50	248,197	26.85%	13%
High confidence	50	141,962	−15.76%	65%
Low confidence	50	222,750	34.58%	5%

5 Frozen causal gate and prospective extension

The audited causal gate used only permitted source-time and causal-history features. An initial leakage failure was a false warning caused by scanning the YAML forbidden-pattern list itself. The corrected audit classified the gate as structurally valid with limitations. It selected B5 for 0% of eligible comparisons, making its predictions identical to always using B3. The gate therefore failed as a prospective selector.

Gate scoring requires the thirteen instrumented source-time features, which the frozen object-level diagnostic table records for YOLO11n and YOLO11s only. Extension flow: YOLO11n had 45 detector-complete, 10 propagation-complete, and 9 gate-eligible logs; YOLO11s had 45 detector-complete, 45 propagation-complete, and 33 gate-eligible logs; RT-DETR-L had 21 propagation-complete logs and no gate-eligible logs.

The B3-versus-B5 contrast requires only predicted and target geometry and was therefore scored on the RT-DETR-L extension logs.

Table 4: Prospective replication at 300 ms in high-yaw extension logs.

Detector	Logs	B3 error	B5 error	Disadvantage (95% CI)
YOLO11n	9	.006143	.008478	29.21% (17.80–40.77%)
YOLO11s	33	.006491	.007269	20.35% (12.20–24.67%)
RT-DETR-L	15	.007574	.008746	10.90% (3.39–31.17%)

6 Detection density and threshold comparability

A common confidence threshold does not equalize output density across architectures, so absolute false-positive counts differ across detectors for reasons preceding propagation. All method contrasts are paired within detector and log.

Table 5: Detector output density at the frozen endpoint, confidence threshold 0.25.

Detector	Frames	Det./frame	Median conf.	Frac. conf. < 0.4
YOLO11n	7,618	7.90	0.521	31.8%
YOLO11s	7,715	9.80	0.606	25.2%
RT-DETR-L	7,719	20.50	0.476	40.6%

Table 6: Class-stratified results at the frozen 300-ms high-yaw endpoint.

Detector	Class	Logs	Objects	B5 disadvantage (95% CI)
YOLO11n	Vehicle	40	10,562	20.48% (11.72–37.21%)
YOLO11n	Ped./cyclist	23	527	2.60% (0.00–16.80%)
YOLO11n	Other	18	336	0.97% (−0.79–4.58%)
YOLO11s	Vehicle	40	12,668	19.01% (10.37–30.08%)
YOLO11s	Ped./cyclist	25	894	16.63% (0.28–65.80%)
YOLO11s	Other	23	527	0.00% (−0.81–1.55%)
RT-DETR-L	Vehicle	40	21,802	15.92% (7.68–22.47%)
RT-DETR-L	Ped./cyclist	29	1,872	26.64% (1.49–44.53%)
RT-DETR-L	Other	28	2,055	1.16% (−0.29–4.28%)

7 Exploratory class-stratified analysis

The analysis was exploratory and used complete logs as the independent units with 10,000 paired log-cluster bootstrap replicates. RT-DETR-L rows were reconstructed from the frozen low-level prediction and target tables; running the same reconstruction on the YOLO detectors reproduced the frozen values to a maximum absolute difference of 1.7×10^{-18} , and per-class comparison counts sum to the held-out totals of 11,425, 14,089, and 25,729.

8 Image-space direction agreement

Table 7: Agreement between the injected correction and the residual it must remove.

Detector	Eligible	Toward	Away	Med. corr. (px)	Med. resid. (px)
YOLO11n	10,839	57.75%	42.25%	13.58	18.10
YOLO11s	13,228	58.03%	41.97%	11.36	17.47
RT-DETR-L	22,735	56.62%	43.38%	8.99	22.39

B5 adds an injected correction to the B3 box, so the displacement that would zero the error is the residual between the B3 center and the target center. Direction is the sign of the inner product between the injected correction and that residual. Objects whose correction or residual fell below one pixel were treated as ambiguous and excluded, comprising 5.1%, 6.1%, and 11.6% of objects. This statistic is defined in the image space where error is scored and is distinct from the ground-plane velocity comparison reported above.

9 Motion-direction agreement

Direction was defined by the sign of the dot product between estimated two-dimensional ground-plane velocity and oracle source-to-target displacement. Objects with true speed below 0.1 m/s were

Table 8: Agreement between estimated ground-plane motion and oracle displacement.

Detector	Eligible	Same half-plane	Opposite half-plane	Ambiguous
YOLO11n	5,783	51.08%	48.92%	45.25%
YOLO11s	7,138	51.41%	48.59%	43.65%

classified as ambiguous and excluded from the same-versus-opposite half-plane percentages.

10 Runtime scope

Median per-object propagation times were 0.2360 ms for YOLO11n B3, 0.2216 ms for YOLO11n B5, 0.2144 ms for YOLO11s B3, 0.2005 ms for YOLO11s B5, 0.3298 ms for RT-DETR-L B3, and 0.3094 ms for RT-DETR-L B5. In a separate 2,000-frame benchmark excluding detector inference and input/output, median total post-detection overhead was 1.26 ms per frame for B3 and 1.14 ms for B5; 95th percentiles were 2.51 and 2.14 ms. Association, velocity-estimation, and gate stages were not timed separately, so the small aggregate difference is not interpreted.

11 Reproducibility files

The Overleaf package includes the aggregate source tables used for the updated figures. The repository remains the authoritative source for low-level outputs, frozen configurations, hashes, and tests.